

Chenxi Feng

Tel: +1 (607) 227-0681; Email: cf545@cornell.edu; Website: <https://fcxee.github.io/>

EDUCATION

Cornell University	Ithaca, United States
<i>Ph.D. in Agronomy and Crop Science</i>	09/2024 - Present
Nanjing University	Nanjing, China
<i>B.S. in Atmospheric Science</i>	09/2020 - 06/2024
Thesis: Improvements and Applications of Methane Retrieval Approach Using Imaging Spectrometer	

RESEARCH INTERESTS

Numerical modeling of land-atmosphere carbon cycling
AI-based data-model fusion
Climatological sensitivity of soil respiration
Methane point source emission retrieval
Aerosol radiative properties

PUBLICATIONS

(* , co-first author)

- 2026 **Feng, C.**, Zhang, S., Qin, S., Xu, X., Tao, F., Luo, Z., ... & Luo, Y. Global convergence in temperature response of microbial respiratory soil organic carbon loss. *In prep.*
- 2026 Li, F.*, **Feng, C.***, Juan, J. R., Guanter, L., Chen, H., Sun, S., ... & Zhang, Y. An optimized methane retrieval approach based on morphological fusion for mapping methane point emissions from spaceborne imaging spectroscopy. *Remote Sensing of Environment*. [10.1016/j.rse.2026.115718](https://doi.org/10.1016/j.rse.2026.115718)
- 2026 Gang, C., Wei, N., **Feng, C.**, Xu, H., Liu, H., Tao, F., ... & Luo, Y. Net primary productivity orchestrates uncertainty sources driving global soil organic carbon under land use change. *Communications Earth & Environment*. [10.1038/s43247-026-03312-6](https://doi.org/10.1038/s43247-026-03312-6)
- 2026 Li, F., Lin, K., Yan, Y., Bai, S., Huang, Q., **Feng, C.**, ... & Zhang, Y. Comparing the performance of different hyperspectral satellite imaging spectroscopy in mapping methane point-source emissions. *Remote Sensing of Environment*. [10.1016/j.rse.2025.115224](https://doi.org/10.1016/j.rse.2025.115224)
- 2024 **Feng, C.**, Chen, S., Zeng, Z.-C., Luo, Y., Natraj, V., & Yung, Y. L. Aerosol-Calibrated matched filter method for retrievals of methane point source emissions over the Los Angeles Basin. *Earth and Space Science*. [10.1029/2024EA003519](https://doi.org/10.1029/2024EA003519)
- 2024 Li, F., Bai, S., Lin, K., **Feng, C.**, Sun, S., Zhao, S., ... & Zhang, Y. Satellite-based surveys reveal substantial methane point-source emissions in major oil & gas basins of North America during 2022-2023. *Journal of Geophysical Research: Atmospheres*. [10.1029/2024JD040870](https://doi.org/10.1029/2024JD040870)

PRESENTATION

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- 2026 **Feng, C.**, Zhang, S., Qin, S., Xu, X., Tao, F., Luo, Z., Yang, Y., Luo, Y. Substrate-induced attenuation of thermal accelerating in soil organic carbon mineralization, 2026 Ecological Society of America Meeting, 26-31 July 2026, Salt Lake City, UT, Poster
- 2024 **Feng, C.**, Chen, S., Zeng, Z.-C., Luo, Y., Natraj, V., & Yung, Y. L. Assessing aerosol-induced bias in methane point source emission retrievals in the LA basin, 2023 American Geophysical Union Fall Meeting, 11-15 December 2023, San Francisco, CA, Poster.

RESEARCH BACKGROUNDS

Cornell University

Ithaca, NY, U.S.

PhD student (advisor: Prof. Yiqi Luo)

Global convergence in responses of soil organic carbon loss to warming

02/2025 – Present

- Performed Bayesian inversion analysis on soil incubation dataset
- Constrained a Community Land Model Version 5 (CLM5)-type soil organic carbon (SOC) cycle model
- Identified convergent underlying kinetics controlling divergent responses of microbial mineralization-induced SOC loss to warming

California Institute of Technology

Pasadena, CA, U.S.

Research Intern (advisor: Prof. Yuk L. Yung)

06/2023 - 01/2024

Assessing and eliminating impacts of aerosol on methane point source emission retrieval

- Simulated radiance spectra with the 2S-ESS RT model to study aerosol-induced bias on methane concentration estimates
- Developed an aerosol-calibrated matched filter method for methane retrieval and optimized retrieval accuracy
- Applied the algorithm to methane point sources in Los Angeles using remote sensing data, indicating emission underestimation and proposing improvements

Nanjing University

Nanjing, China

Research Assistant (advisor: Prof. Yongguang Zhang)

02/2022 - 07/2024

Atmospheric methane inversion and point source quantification using hyperspectral observation

- Enhanced methane retrieval accuracy by combining results from weak and strong absorption bands
- Conducted end-to-end simulations to validate proposed methods using ground measured data
- Processed remote sensing data from Chinese satellites to retrieve methane point source emissions
- Analyzed methane plumes in key regions, detecting leaks from oil and gas industries in the US and Middle East

Nanjing University

Nanjing, China

Research Assistant (advisor: Prof. Huilin Chen)

12/2022 - 12/2023

In-situ detection of atmospheric methane and carbon dioxide using unmanned aerial vehicle

- Designed an unmanned aircraft methane and carbon dioxide monitoring system using Axetris AG and K96 sensors
 - Conducted lab experiments to assess sensor performance, including sensitivity, accuracy, and response time to gases
 - Assisted in planning and preparing field experiments over methane emitters in Nanjing City
 - Analyzed methane plumes using the Gaussian model and estimated methane emission flux rate
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ACHIEVEMENTS AND AWARDS

Outstanding Thesis Award (The First Prize), Nanjing University	2024
Chenxue Scholarship for Overseas Study, Nanjing University	2024
Scholarship of Academic Excellence (The First Prize, Top 5%), Nanjing University	2023
Zhenggang Fellowship for Oversea Studies of Nanjing University (Top 5%), Nanjing University	2022
Scholarship of Academic Excellence (The Second Prize, Top 10%), Nanjing University	2022
Outstanding Student for Theory and Methods of Land Surface Remote Sensing Inversion Summer School 2022, Beijing Normal University	2022
Scholarship of Academic Excellence (The Second Prize, Top 10%), Nanjing University	2021

PROGRAMMING SKILLS

Python, MATLAB, R, FORTRAN